

The Derivatives

Definition

If the function $y = f(x)$ is **continuous at point x** , the following limit can be calculated:

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This limit is denoted by $\frac{dy}{dx}$ or y' or f' and is called the **first derivative** of the function $y = f(x)$.

Example

Using the definition (**first principles**), find derivative of following functions

(1) $y = x^2 + 1$

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{((x+h)^2 + 1) - (x^2 + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + (h)^2 + 1) - (x^2 + 1)}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} = 2x \end{aligned}$$

(2) $y = \sqrt{x}$

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{\sqrt{x+h} - \sqrt{x}}{h} \right) \left(\frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \right) = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} = \frac{1}{2\sqrt{x}} \end{aligned}$$

(3) $y = c$; where c is constant

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0$$

Techniques of finding derivatives

(1) Derivative of Sum or difference Rule

If $y = u(x) \pm v(x)$
then $\frac{dy}{dx} = \frac{du}{dx} \pm \frac{dv}{dx}$

(2) Derivative of Product Rule

If $y = u(x) \cdot v(x)$
then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

(3) Derivative of Quotient Rule

If $y = \frac{u(x)}{v(x)}$
then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

(4) The Chain Rule

If $y = y(u)$ and $u = u(x)$
then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

(5) Derivative of Power Rule

If $y = x^n$

then $\frac{dy}{dx} = nx^{n-1}$

In general

✱ If $y = [f(x)]^n$
then $\frac{dy}{dx} = n[f(x)]^{n-1}f'(x)$

✱ If $y = \sqrt{f(x)}$
then $\frac{dy}{dx} = \frac{f'(x)}{2\sqrt{f(x)}}$

✱ If $y = \frac{1}{f(x)}$
then $\frac{dy}{dx} = \frac{-f'(x)}{[f(x)]^2}$

Example: find $\frac{dy}{dx}$ in the following

(1) $y = (2x^2 + 1)^{20}$

$$\frac{dy}{dx} = 20(2x^2 + 1)^{19}(2x^2 + 1)' = 80x(2x^2 + 1)^{19}$$

(2) $y = \sqrt{x^3 - 2x}$

$$\frac{dy}{dx} = \frac{(x^3 - 2x)'}{2\sqrt{x^3 - 2x}} = \frac{3x^2 - 2}{2\sqrt{x^3 - 2x}}$$

(3) $y = \left(\frac{1}{x+1}\right)^4$

$$\frac{dy}{dx} = 4\left(\frac{1}{x+1}\right)^3 \left(\frac{1}{x+1}\right)' = 4\left(\frac{1}{x+1}\right)^3 \left(\frac{-1}{(x+1)^2}\right) = \frac{-4}{(x+1)^5}$$

(6) Derivative of Trigonometric Functions

(1) If $y = \sin(x)$

then $\frac{dy}{dx} = \cos(x)$

(2) If $y = \cos(x)$

then $\frac{dy}{dx} = -\sin(x)$

In general

✱ If $y = \sin(f(x))$
then $\frac{dy}{dx} = f'(x) \cdot \cos(f(x))$

✱ If $y = \cos(f(x))$
then $\frac{dy}{dx} = -f'(x) \cdot \sin(f(x))$

Example: find $\frac{dy}{dx}$ in the following

(1) $y = \sin(3x)$

$$\frac{dy}{dx} = (3x)' \cos(3x) = 3 \cos(3x)$$

(2) $y = \sin(x^2 - x)$

$$\frac{dy}{dx} = (x^2 - x)' \cos(x^2 - x) = (2x - 1) \cos(x^2 - x)$$

(3) $y = \sin(\sqrt{x})$

$$\frac{dy}{dx} = (\sqrt{x})' \cos(\sqrt{x}) = \frac{\cos(\sqrt{x})}{2\sqrt{x}}$$

Example: find $\frac{dy}{dx}$ in the following

(1) $y = \cos(3x - 1)$

$$\frac{dy}{dx} = -(3x - 1)' \sin(3x - 1) = -3 \sin(3x - 1)$$

(2) $y = \cos(1 - x^2)$

$$\frac{dy}{dx} = -(1 - x^2)' \sin(1 - x^2) = (2x) \sin(1 - x^2)$$

Note

By using the derivative of $\sin(x)$ and $\cos(x)$, we can obtain **the derivative of others trigonometric functions** such as $\tan(x)$, $\sec(x)$ and $\operatorname{cosec}(x)$

(1) $y = \tan(x) = \frac{\sin(x)}{\cos(x)}$

$$\frac{dy}{dx} = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x)$$

In general

✱ If $y = \tan(f(x))$

then $\frac{dy}{dx} = f'(x) \cdot \sec^2(f(x))$

Example: find $\frac{dy}{dx}$ in the following

(1) $y = \sec(x)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{1}{\cos(x)} \right] = \frac{\cos(x)(0) - (1)(-\sin(x))}{\cos^2(x)} \\ &= \frac{\sin(x)}{\cos^2(x)} = \frac{1}{\cos(x)} \frac{\sin(x)}{\cos(x)} = \sec(x) \tan(x) \end{aligned}$$

(2) $y = \operatorname{cosec}(x)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{1}{\sin(x)} \right] = \frac{\sin(x)(0) - (1)(\cos(x))}{\sin^2(x)} \\ &= \frac{-\cos(x)}{\sin^2(x)} = \frac{-1}{\sin(x)} \frac{\cos(x)}{\sin(x)} = -\operatorname{cosec}(x) \cot(x) \end{aligned}$$

(3) $y = \cot(x)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{\cos(x)}{\sin(x)} \right] = \frac{\sin(x)(-\sin(x)) - (\cos(x))(\cos(x))}{\sin^2(x)} \\ &= \frac{-1}{\sin^2(x)} = -\sec^2(x) \end{aligned}$$

Exercise (6-1)

1- Using the definition (first principles), find derivative of following functions

$$(1) y = x^3 \quad (2) y = \frac{1}{x} \quad (3) y = \sqrt{x-1}$$

$$(4) y = \frac{1}{\sqrt{x}} \quad (5) y = 2x^4 - x \quad (6) y = 5 - x$$

2- find $\frac{dy}{dx}$ in the following

$$(1) y = x^4 + 3x^2 - 6 \quad (2) y = 6x^3 - x^2 \quad (3) y = \frac{x^2-3}{a+b}$$

$$(4) y = \frac{x^3+2x^2-3}{3} \quad (5) y = 2ax^3 - \frac{x^2}{b} + 3 \quad (6) y = \frac{(x+1)^3}{x^2}$$

$$(7) y = 6x^{\frac{7}{2}} + 4x^{\frac{5}{2}} + 2x \quad (8) y = \sqrt{3x} + \sqrt[3]{x} + \frac{1}{x} \quad (9) y = \sqrt[3]{x^2}$$

$$(10) y = \frac{ax^2}{\sqrt[3]{x}} + \frac{b}{x\sqrt{x}} - \frac{\sqrt[3]{x}}{\sqrt{x}} \quad (11) y = (1+4x^3)(1+2x^2) \quad (12) y = \frac{2x^4}{b^2-x^2}$$

$$(13) y = \frac{a-x}{a+x} \quad (14) y = \frac{x^3}{1+x^2} \quad (15) y = \frac{(x+4)^2}{x+3}$$

$$(16) y = \frac{x^3+2}{x^2-x-2} \quad (17) y = \frac{x^p}{x^m-a^m} \quad (18) y = (a^2 - x^2)^4$$

$$(19) y = \sqrt{a^2 + x^2} \quad (20) y = (a+x)\sqrt{a-x} \quad (21) y = \sqrt{\frac{1+x}{1-x}}$$

$$(22) y = \frac{2x^2-1}{x\sqrt{1+x^2}} \quad (23) y = (1 + \sqrt[3]{x})^3 \quad (24) y = \tan(x+3)$$

$$(25) y = \frac{\sin(x)}{1+\cos(x)} \quad (26) y = \sin(2x)\cos(3x) \quad (27) y = 3\sqrt{\cos(2x)}$$

$$(28) y = \frac{\tan(\frac{x}{2}) + \cot(\frac{x}{2})}{x} \quad (29) y = \sin^2(x)\cos^4(4x) \quad (30) y = \left(\frac{1+\sin(x)}{1-\sin(x)}\right)^{\frac{1}{2}}$$